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```
MariaDB [(none)]> create database work;  
Query OK, 1 row affected (0.016 sec)
```

```
MariaDB [(none)]> use work;  
Database changed
```

--Creating tables and adding the constraint

```
MariaDB [work]> create table employee(empNo varchar(10) primary key not null, fName  
varchar(20) not null, lName varchar(20) not null, address varchar(30) not null, DOB  
date not null, sex char(1) not null, position varchar(20), deptNo int not null);  
Query OK, 0 rows affected (0.029 sec)
```

```
MariaDB [work]> alter table employee add constraint chk_sex check(sex in('M','F'));  
Query OK, 0 rows affected (0.039 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [work]> alter table employee add constraint chk_pos check(position in  
( 'Manager', 'Team Leader', 'Analyst', 'Software Developer' ));  
Query OK, 0 rows affected (0.050 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [work]> create table department(deptNo varchar(20) primary key not null,  
deptName varchar(20) not null, mrgEmpNo varchar(20) not null, foreign key(mrgEmpNo)  
references employee(empNo));  
Query OK, 0 rows affected (0.031 sec)
```

```
MariaDB [work]> create table project(projNo varchar(20) primary key not null,  
projName varchar(30) not null, deptNo varchar(20) not null, foreign key(deptNo)  
references department(deptNo));  
Query OK, 0 rows affected (0.016 sec)
```

```
MariaDB [work]> create table worksOn(empNo varchar(20) not null references  
employee(empNo), projNo varchar(20) not null references project(projNo), dateWorked  
date not null, hoursWorked int not null, primary key(empNo,projNo,dateWorked));  
Query OK, 0 rows affected (0.011 sec)
```

```
MariaDB [work]> alter table worksOn modify column hoursWorked int not null, add  
constraint chk_hoursWorked check(hoursWorked between 0 and 40);  
Query OK, 0 rows affected (0.048 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [work]> alter table employee modify column deptNo varchar(20);  
Query OK, 0 rows affected (0.054 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

MariaDB [work]> describe employee;

Field	Type	Null	Key	Default	Extra
empNo	varchar(10)	NO	PRI	NULL	
fName	varchar(20)	NO		NULL	
lName	varchar(20)	NO		NULL	
address	varchar(30)	NO		NULL	
DOB	date	NO		NULL	
sex	char(1)	NO		NULL	
position	varchar(20)	YES		NULL	
deptNo	varchar(20)	YES		NULL	

8 rows in set (0.004 sec)

MariaDB [work]> describe department;

Field	Type	Null	Key	Default	Extra
deptNo	varchar(20)	NO	PRI	NULL	
deptName	varchar(20)	NO		NULL	
mrgEmpNo	varchar(20)	NO	MUL	NULL	

3 rows in set (0.009 sec)

MariaDB [work]> describe project;

Field	Type	Null	Key	Default	Extra
projNo	varchar(20)	NO	PRI	NULL	
projName	varchar(30)	NO		NULL	
deptNo	varchar(20)	NO	MUL	NULL	

3 rows in set (0.025 sec)

MariaDB [work]> describe worksOn;

Field	Type	Null	Key	Default	Extra
empNo	varchar(20)	NO	PRI	NULL	
projNo	varchar(20)	NO	PRI	NULL	
dateWorked	date	NO	PRI	NULL	
hoursWorked	int(11)	NO		NULL	

4 rows in set (0.007 sec)

MariaDB [work]> INSERT INTO employee (empNo, fName, lName, address, DOB, sex, position, deptNo) VALUES

-> ('E001', 'Alice', 'Johnson', '123 Main St', '1985-03-15', 'F', 'Manager', 'D001'),

```

-> ('E002', 'Bob', 'Smith', '456 Oak Ave', '1982-07-22', 'M', 'Manager',
'D002'),
-> ('E003', 'Carol', 'Williams', '789 Pine Rd', '1990-11-05', 'F', 'Manager',
'D003'),
-> ('E004', 'David', 'Brown', '321 Elm St', '1988-04-12', 'M', 'Manager',
'D004'),
-> ('E005', 'Emma', 'Davis', '654 Maple Dr', '1992-09-30', 'F', 'Manager',
'D005'),
-> ('E006', 'Frank', 'Wilson', '987 Cedar Ln', '1995-01-18', 'M', 'Software
Developer', 'D003'),
-> ('E007', 'Grace', 'Taylor', '135 Birch Blvd', '1993-06-25', 'F', 'Analyst',
'D002'),
-> ('E008', 'Henry', 'Anderson', '246 Spruce Way', '1991-12-10', 'M', 'Team
Leader', 'D003'),
-> ('E009', 'Ivy', 'Thomas', '357 Walnut Ct', '1994-08-14', 'F', 'Software
Developer', 'D003'),
-> ('E010', 'Jack', 'Martinez', '468 Ash Rd', '1987-05-03', 'M', 'Analyst',
'D004');
Query OK, 10 rows affected (0.005 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```

MariaDB [work]> INSERT INTO department (deptNo, deptName, mrgEmpNo) VALUES
-> ('D001', 'Human Resources', 'E001'),
-> ('D002', 'Finance', 'E002'),
-> ('D003', 'IT', 'E003'),
-> ('D004', 'Marketing', 'E004'),
-> ('D005', 'Operations', 'E005');
Query OK, 5 rows affected (0.001 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

```

MariaDB [work]> INSERT INTO project (projNo, projName, deptNo) VALUES
-> ('P001', 'Employee Portal Upgrade', 'D003'),
-> ('P002', 'Financial Reporting System', 'D002'),
-> ('P003', 'Marketing Campaign 2026', 'D004'),
-> ('P004', 'HR Payroll Automation', 'D001'),
-> ('P005', 'Network Infrastructure Refresh', 'D003'),
-> ('P006', 'Customer Analytics Dashboard', 'D004');
Query OK, 6 rows affected (0.020 sec)
Records: 6 Duplicates: 0 Warnings: 0

```

```

MariaDB [work]> INSERT INTO worksOn (empNo, projNo, dateWorked, hoursWorked) VALUES
-> ('E006', 'P001', '2026-06-01', 8),
-> ('E008', 'P001', '2026-06-01', 7),
-> ('E009', 'P001', '2026-06-02', 6),
-> ('E007', 'P002', '2026-06-01', 5),
-> ('E007', 'P002', '2026-06-02', 8),
-> ('E010', 'P003', '2026-06-03', 4),
-> ('E006', 'P005', '2026-06-03', 9),
-> ('E008', 'P005', '2026-06-04', 8),
-> ('E009', 'P001', '2026-06-04', 7),

```

```

-> ('E003', 'P001', '2026-06-05', 3),
-> ('E001', 'P004', '2026-06-05', 6),
-> ('E004', 'P006', '2026-06-06', 5);

```

Query OK, 12 rows affected (0.020 sec)
Records: 12 Duplicates: 0 Warnings: 0

MariaDB [work]> select * from employee;

```

+-----+-----+-----+-----+-----+-----+-----+
+-----+
| empNo | fName | lName   | address          | DOB          | sex | position
| deptNo |
+-----+-----+-----+-----+-----+-----+-----+
+-----+
| E001  | Alice | Johnson | 123 Main St     | 1985-03-15  | F   | Manager
| D001  |
| E002  | Bob   | Smith   | 456 Oak Ave     | 1982-07-22  | M   | Manager
| D002  |
| E003  | Carol | Williams| 789 Pine Rd     | 1990-11-05  | F   | Manager
| D003  |
| E004  | David | Brown   | 321 Elm St      | 1988-04-12  | M   | Manager
| D004  |
| E005  | Emma  | Davis   | 654 Maple Dr    | 1992-09-30  | F   | Manager
| D005  |
| E006  | Frank | Wilson  | 987 Cedar Ln    | 1995-01-18  | M   | Software Developer
| D003  |
| E007  | Grace | Taylor  | 135 Birch Blvd  | 1993-06-25  | F   | Analyst
| D002  |
| E008  | Henry | Anderson| 246 Spruce Way  | 1991-12-10  | M   | Team Leader
| D003  |
| E009  | Ivy   | Thomas  | 357 Walnut Ct   | 1994-08-14  | F   | Software Developer
| D003  |
| E010  | Jack  | Martinez| 468 Ash Rd      | 1987-05-03  | M   | Analyst
| D004  |
+-----+-----+-----+-----+-----+-----+-----+
+-----+
10 rows in set (0.000 sec)

```

MariaDB [work]> select * from department;

```

+-----+-----+-----+
| deptNo | deptName          | mrgEmpNo |
+-----+-----+-----+
| D001   | Human Resources   | E001     |
| D002   | Finance           | E002     |
| D003   | IT                | E003     |
| D004   | Marketing          | E004     |
| D005   | Operations        | E005     |
+-----+-----+-----+
5 rows in set (0.001 sec)

```

MariaDB [work]> select * from project;

projNo	projName	deptNo
P001	Employee Portal Upgrade	D003
P002	Financial Reporting System	D002
P003	Marketing Campaign 2026	D004
P004	HR Payroll Automation	D001
P005	Network Infrastructure Refresh	D003
P006	Customer Analytics Dashboard	D004

6 rows in set (0.001 sec)

MariaDB [work]> select * from worksOn;

empNo	projNo	dateWorked	hoursWorked
E001	P004	2026-06-05	6
E003	P001	2026-06-05	3
E004	P006	2026-06-06	5
E006	P001	2026-06-01	8
E006	P005	2026-06-03	9
E007	P002	2026-06-01	5
E007	P002	2026-06-02	8
E008	P001	2026-06-01	7
E008	P005	2026-06-04	8
E009	P001	2026-06-02	6
E009	P001	2026-06-04	7
E010	P003	2026-06-03	4

12 rows in set (0.000 sec)

--Question 7.25

MariaDB [work]> create view female_manager_projects as select
p.projNo,p.projName,p.deptNo from project p join department d on p.deptNo=d.deptNo
join employee e on d.mrgEmpNo=e.empNo where e.sex='F' and e.position='Manager' order
by p.projNo;
Query OK, 0 rows affected (0.026 sec)

MariaDB [work]> select * from female_manager_projects;

projNo	projName	deptNo
P001	Employee Portal Upgrade	D003
P004	HR Payroll Automation	D001
P005	Network Infrastructure Refresh	D003

3 rows in set (0.001 sec)

--Question 7.26

```
MariaDB [work]> create view employee_description as select e.empNo,
e.fName,e.lName,p.projName,w.hoursWorked from employee e join project p on
p.deptNo=e.deptNo join worksOn w on w.empNo=e.empNo order by e.empNo;
Query OK, 0 rows affected (0.023 sec)
```

```
MariaDB [work]> select * from employee_description;
```

empNo	fName	lName	projName	hoursWorked
E001	Alice	Johnson	HR Payroll Automation	6
E003	Carol	Williams	Employee Portal Upgrade	3
E003	Carol	Williams	Network Infrastructure Refresh	3
E004	David	Brown	Marketing Campaign 2026	5
E004	David	Brown	Customer Analytics Dashboard	5
E006	Frank	Wilson	Employee Portal Upgrade	8
E006	Frank	Wilson	Network Infrastructure Refresh	8
E006	Frank	Wilson	Employee Portal Upgrade	9
E006	Frank	Wilson	Network Infrastructure Refresh	9
E007	Grace	Taylor	Financial Reporting System	5
E007	Grace	Taylor	Financial Reporting System	8
E008	Henry	Anderson	Employee Portal Upgrade	7
E008	Henry	Anderson	Network Infrastructure Refresh	7
E008	Henry	Anderson	Employee Portal Upgrade	8
E008	Henry	Anderson	Network Infrastructure Refresh	8
E009	Ivy	Thomas	Employee Portal Upgrade	6
E009	Ivy	Thomas	Network Infrastructure Refresh	6
E009	Ivy	Thomas	Employee Portal Upgrade	7
E009	Ivy	Thomas	Network Infrastructure Refresh	7
E010	Jack	Martinez	Marketing Campaign 2026	4
E010	Jack	Martinez	Customer Analytics Dashboard	4

21 rows in set (0.021 sec)

--Question 7.27

```
MariaDB [work]> create view EmpProject(empNo, projNo, totalHours) as select w.empNo,
w.projNo, sum(w.hoursWorked) as totalHours from WorksOn w join Employee e on e.empNo
= w.empNo join Project p on p.projNo = w.projNo group by w.empNo, w.projNo;
Query OK, 0 rows affected (0.007 sec)
```

--Question 7.27 a)

```
MariaDB [work]> select * from EmpProject;
```

empNo	projNo	totalHours
E001	P004	6
E003	P001	3
E004	P006	5
E006	P001	8

E006	P005	9
E007	P002	13
E008	P001	7
E008	P005	8
E009	P001	13
E010	P003	4

+-----+-----+
10 rows in set (0.001 sec)

--Question 7.27 b)

MariaDB [work]> select projNo from EmpProject where projNo='SCCS';
Empty set (0.001 sec)

--Quetion 7.27 c)

MariaDB [work]> select count(projNo) from EmpProject where empNo='E1';

count(projNo)
0

+-----+
1 row in set (0.001 sec)

MariaDB [work]> select empNo, totalHours from EmpProject group by empNo;

empNo	totalHours
E001	6
E003	3
E004	5
E006	8
E007	13
E008	7
E009	13
E010	4

+-----+
8 rows in set (0.002 sec)

General

MariaDB [work]> create table part(partNo varchar(10), contract varchar(10), partCost decimal(8,2), primary key(partNo, contract));
Query OK, 0 rows affected (0.014 sec)

MariaDB [work]> describe part;

Field	Type	Null	Key	Default	Extra
-------	------	------	-----	---------	-------

partNo	varchar(10)	NO	PRI	NULL		
contract	varchar(10)	NO	PRI	NULL		
partCost	decimal(8,2)	YES		NULL		

3 rows in set (0.021 sec)

```
MariaDB [work]> INSERT INTO part (partNo, contract, partCost) VALUES
-> ('P001', 'C100', 125.50),
-> ('P001', 'C101', 132.75),
-> ('P002', 'C100', 89.99),
-> ('P002', 'C102', 95.25),
-> ('P003', 'C103', 245.00),
-> ('P004', 'C100', 67.80),
-> ('P004', 'C104', 72.15),
-> ('P005', 'C101', 189.99),
-> ('P006', 'C105', 45.50),
-> ('P007', 'C100', 310.00);
```

Query OK, 10 rows affected (0.002 sec)
Records: 10 Duplicates: 0 Warnings: 0

```
MariaDB [work]> select * from part;
```

partNo	contract	partCost
P001	C100	125.50
P001	C101	132.75
P002	C100	89.99
P002	C102	95.25
P003	C103	245.00
P004	C100	67.80
P004	C104	72.15
P005	C101	189.99
P006	C105	45.50
P007	C100	310.00

10 rows in set (0.001 sec)

```
MariaDB [work]> create view ExpensiveParts(partNo) as select distinct partNo from
part where partCost>1000;
Query OK, 0 rows affected (0.008 sec)
```

```
MariaDB [work]> select * from ExpensiveParts;
Empty set (0.017 sec)
```

General Question

A materialized view stores the results of a query rather than just the query itself. As a result, it must be updated whenever the Part table changes. When a new row is inserted with a partCost greater than 1000, the corresponding partNo is added to the

materialized view. If a row is updated and its partCost increases above 1000, the partNo is also added. However, if the cost drops to 1000 or below, the database must first check whether the same partNo has any other contracts with a partCost greater than 1000 before removing it from the view. Likewise, when a row is deleted, the partNo is removed only if there are no remaining contracts for that part with a cost above 1000.

In some cases, the materialized view can be updated without consulting the Part table. For example, when a newly inserted row has a partCost greater than 1000, the partNo can be added directly because the change cannot affect other records. However, when a row with a partCost above 1000 is updated or deleted, the Part table must be checked to determine whether another qualifying contract for the same partNo still exists before removing it from the materialized view.